

Remark: The following questions don't appear numbered in the scaling book. I will number them in order of appearance as 0.X where  $X \in \mathbb{N}$  and  $X \geq 1$ .

### Question 0.1

Let  $M = 70.4 \text{ e } 9$ . We know that

$$\text{Transformer FLOPs} \simeq 6 \cdot BT \cdot M$$

$$\text{therefore per token we know its GMFLOPs} = \\ = 422.4 \text{ e } 9$$

### Question 0.2

Let  $M = 70.4 \text{ e } 9$  and  $BT = 15 \text{ e } 12$ , hence:

$$\begin{aligned} \text{Transformer FLOPs} &= 6 \cdot BT \cdot M = (6 \cdot 70.4 \cdot 15) \cdot 10^{21} \\ &= 6336 \cdot 10^{21} \simeq 6.3 \text{ e } 24 = 6.3 \text{ yotta FLOPs.} \end{aligned}$$

### Question 0.3

Assume TPU v5p pod of  $16 \times 20 \times 28 = 8960$  chips. Let v5p bf16 FLOPS/s =  $4.59 \times 10^{14}$ .

We know that FLOPS Per Chip =  $\frac{1}{8960} \cdot 6 \cdot BT \cdot M$ .

Let  $M = 70.4 \times 10^9$  and  $BT = 15 \times 10^{12}$ . Therefore

$$\begin{aligned} \text{Time} &= \frac{6 \cdot BT \cdot M}{8960 \cdot 0.4 \cdot 4.59 \times 10^{14}} \\ &= 44.3 \text{ days.} \end{aligned}$$

### Question 0.4

Assume we checkpoint the three big matmuls in the MLP part and

$A = \text{softmax}(Q \cdot K^T) \cdot V$  which has shape  $A[B, T, K, G, H]$  so it occupies  $2 \cdot B \cdot T \cdot K \cdot G \cdot H = 2BTD$

hence, combining our formula from exercise 5.2 we get that:

$$\text{Activation checkpointing} = L \cdot (4BT_F + 4BTD)$$

therefore total memory required is:

$$10 \cdot M + L \cdot (BT) \cdot (4F + 4D)$$

where  $M$  is number of model parameters.

let  $BT = 4e6$ , therefore for LLaMA 3-70B we get:

$$\begin{aligned} & 10 \cdot (70.4e9) + 80 \cdot (4e6) \cdot (4 \cdot 28672 + 4 \cdot 8192) \\ & \simeq 47.8e12 = 47.8 \text{ TB} \end{aligned}$$

so assuming v5p TPU which has HBM Capacity = 96 GB we need at least:

$$\frac{47.8e12}{96e9} \simeq 498$$

TPUs.

### Question 0.5

$$\text{Memory Per Chip} = \frac{47.8 \times 10^{12}}{8960} \text{ Bytes} =$$

$$\approx 5.3 \times 10^9 \text{ Bytes} = 5.3 \text{ GB}.$$

### Question 0.6

Modeling only the MLP part and using sharding strategy from [Sequence Parallelism. Shenggui Li<sup>1</sup> et al] both FSDP and Sequence Parallelism perform:

$$\text{In } [B_x, L_y, D] \rightarrow \text{Win} [D_x, F] \rightarrow \text{Wout} [F, D_x]$$

So without sequence parallelism we get:

$$\text{In } [B_x, L, D] \rightarrow \text{Win} [D_x, F] \rightarrow \text{Wout} [F, D_x].$$

hence :

$$T_{\text{-math}} = \frac{2 \cdot 2 \cdot B \cdot L \cdot D \cdot F}{X \cdot C}$$

$$T_{\text{-wmmms}} = \frac{2 \cdot 2 \cdot D \cdot F}{W \cdot C_I}$$

therefore for being compute bound in this setting we need:

$$\frac{4 B L D F}{4 X D F} > 2550 \Leftrightarrow \frac{B L}{X} > 2550$$

$$\Leftrightarrow \frac{B}{X} > \frac{2550}{L}$$

From the exercise we know that  $L = 4046$ , therefore:

$$\frac{B}{X} > 0.62$$

since  $\frac{B}{X} = 0.11$  we cannot train with

FSOP alone.

### Question 0.7

Again modeling only the MLP part we can see that:

$$\text{In} [B, L, D] \cdot D \cdot \text{Win} [D, F] \cdot F \cdot \text{Wout} [F, D]$$

is equivalent to:

$$\text{In} [B \cdot L, D] \cdot D \cdot \text{Win} [D, F] \cdot F \cdot \text{Wout} [F, D]$$

since no attention involved we just have to reshape after finishing. Therefore with FSOP we will do:

$$\text{In} [B \cdot L_x, D] \cdot D \cdot \text{Win} [D_x, F] \cdot \text{Wout} [F, D]$$

hence the condition to be compute bound is:

$$\frac{(B \cdot L)}{x} > 2550$$

since  $\frac{1024 \cdot 4096}{8960} \approx 468$ , we

cannot use FSDP if we want to be compute bound.

### Question 0.8

We require  $\frac{B \cdot L}{N} > \frac{\alpha^2}{M_x M_y \cdot F}$  and

$X$  to be  $X_{\text{opt}} = \sqrt{\frac{B \cdot L}{F} \cdot \frac{M_x}{M_y} \cdot N}$ , to

be compute bound. Since  $\frac{BL}{N} \approx 468$  and

$\frac{\alpha^2}{M_x M_y F} \approx 113$  if we choose  $X = 2240$ ,

since is close to our  $X_{\text{opt}}$ , we will be compute bound.

## Question 1:

I need the per pod time to be compute bound, and we can use either FSDP or FSDP + TP. We can use Data Parallel over the DCN network and FSDP + TP. So let  $B = N \cdot B_{\text{local}}$  where  $N \in \mathbb{N}$

and reshape  $I_n [B, D]$  to

$I_n [N, B_{\text{local}}, D]$

so the sharding strategy will be:

$$I_n [N_{\text{DCN}}, B_{\text{local}_x}, D_x] \cdot W_{in} [D_x, F_y] \cdot F \cdot W_{out} [F_y, D_x]$$

Note that  $T = \max(T_{\text{pod}}, T_{\text{comms-dcn}})$

where  $T_{\text{pod}} = \max(T_{\text{math}}, T_{\text{comms}})$ .

To be compute bound we need:

$$T_{\text{comms-dcn}} < T_{\text{pod}} \wedge T_{\text{math}} > T_{\text{comms}} \\ \Rightarrow T_{\text{math}} > T_{\text{comms-dcn}}.$$

$$T_{\text{comms-dcn}} = \frac{8 D F}{|X| \cdot |Y|}$$

Assume  $T_{\text{math}} > T_{\text{comms}}$  and note that in the backward pass:

$$T_{\text{math}} = \frac{N}{\text{DCN}} \cdot \frac{\text{BLOCAL}}{|X|} \cdot \frac{D \cdot F}{|Y|} \cdot 2 \cdot 4$$

$$= \frac{8 B D F}{\text{DCN} \cdot |X| \cdot |Y|}$$

Therefore if  $T_{\text{math}} > T_{\text{comms}}$ , the first condition to be compute bound is satisfied if:

$$\frac{800F}{DCN \cdot |X| \cdot |Y|} \cdot \frac{|X| \cdot |Y|}{80F} > \frac{C}{W_{DCN}} = 73.440$$

$$\Leftrightarrow \frac{B}{DCN} > 73440$$

where DCN is the number of pods connected by DCN.

Now to satisfy condition 2 we need as in the exercise before:

$$\frac{B_{LOCAL}}{|X| \cdot |Y|} > \frac{\alpha^2}{M_x M_y \cdot F} \simeq 113$$

and choose  $X$  integer close to  $X_{opt}$  as defined before.

Since  $B = N \cdot B_{LOCAL}$  and  $N \geq DCN$  we get :

$$B_{LOCAL} > |X| \cdot |Y| \cdot 113$$

$$\Rightarrow N \cdot B_{LOCAL} > N \cdot |X| \cdot |Y| \cdot 113 \geq DCN \cdot |X| \cdot |Y| \cdot 113$$

$$\Rightarrow B \geq DCN \cdot |X| \cdot |Y| \cdot 113$$

So if we choose  $B = 426$  and  $N = 4$  we satisfy the conditions.

The time required for training is roughly (assuming 40% MFC in bfloat16)

$$\frac{6 \cdot 10^6 \cdot 70^4}{8960 \cdot 0.4 \cdot 4.59 \cdot 10^{14}} = 255 \text{ ms}$$

per 4M batch step. Therefore

$$\frac{15 \cdot 10^{12}}{4 \cdot 10^6} \cdot 255 \cdot 10^{-3} \approx 11 \text{ days.}$$

Question 2.a:

$$L = 126$$

$$V = 128256$$

$$D = 16384$$

$$F = 53248$$

$$N = 128$$

$$K = 8$$

$$H = 128$$

Using formulas from previous section  
 since architecture does not change just its  
 hyperparams we get:

$$\text{Parameters} = 405e9$$

$$\text{FLOPS Per Step} = 6 \cdot 405e9 = 2.4 \text{ TFLOPS}$$

$$\begin{aligned} \text{FLOPS Total} &= 6 \cdot 15e12 \cdot 405e9 = 3.645e25 \\ &= 36.45 \text{ Volta FLOPS} \end{aligned}$$

### Question 2.b

Using our derivation from question 1,  
 we can use that same sharding strategy  
 and be compute bound if:

$$\frac{B}{DCN} > 73440 \wedge B \geq DCN \cdot 181 \cdot 141 \cdot 113 \wedge N = DCN \wedge B_{local} = \frac{B}{N}$$

Choosing  $B = 16e6$  so that we are safely in  
 the compute bound, we get  $B_{local} = 2e6$ , and

and choosing  $X$  integer close to

$X_{\text{opt}}$ . In this case  $X=1120$  and  $Y=8$ .

A rough estimate for training time would be:

$$\frac{6 \cdot 2e6 \cdot 405e9}{8960 \cdot 0.4 \cdot 4.59e14} \cdot \frac{18e12}{16e6} \approx 32 \text{ days}.$$